

Ex. No.: 11

DATABASE WITH AUTO_INCREMENT SEQUENCES

1)

```
mysql> SELECT * FROM DBMS_Stud;
+-----+-----+-----+
| Register_No | Name | Department |
+-----+-----+-----+
|          1 | Arun | CSE        |
|          2 | Bala | ECE        |
|          3 | Divya | CSE        |
|          4 | Meena | IT         |
+-----+-----+-----+
4 rows in set (0.098 sec)
```

2)

```
mysql> SELECT * FROM DBMS_Stud;
+-----+-----+-----+
| Register_No | Name | Department |
+-----+-----+-----+
|          1 | Arun | CSE        |
|          2 | Bala | ECE        |
|          3 | Divya | CSE        |
|          4 | Meena | IT         |
|         10 | Kiran | IT         |
+-----+-----+-----+
5 rows in set (0.008 sec)
```

3)

```
DBMS_Stud | CREATE TABLE `dbms_stud` (
  `Register_No` int NOT NULL AUTO_INCREMENT,
  `Name` varchar(30) DEFAULT NULL,
  `Department` varchar(20) DEFAULT NULL,
  PRIMARY KEY (`Register_No`)
ENGINE=InnoDB AUTO_INCREMENT=11 DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci |
```

4)

```
1 row in set (0.090 sec)
```

```
mysql> ALTER TABLE DBMS_Stud  
      -> MODIFY Register_No INT PRIMARY KEY;  
ERROR 1068 (42000): Multiple primary key defined  
mysql> SHOW CREATE TABLE DBMS_Stud;
```

5)

```
| DBMS_Stud | CREATE TABLE `dbms_stud` (  
  `Register_No` int NOT NULL AUTO_INCREMENT,  
  `Name` varchar(30) DEFAULT NULL,  
  `Department` varchar(20) DEFAULT NULL,  
  PRIMARY KEY (`Register_No`)  
) ENGINE=InnoDB AUTO_INCREMENT=11 DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci |
```

6)

```
mysql> SELECT * FROM DBMS_Stud;  
+-----+-----+-----+  
| Register_No | Name  | Department |  
+-----+-----+-----+  
|          1 | Arun  | CSE        |  
|          2 | Bala  | ECE        |  
|          3 | Divya | CSE        |  
|          4 | Meena | IT         |  
|         10 | Kiran | IT         |  
|         11 | Ravi  | CSE        |  
+-----+-----+-----+  
6 rows in set (0.162 sec)
```

Ex:No: 12

Simple Programming using REPEAT, WHILE

1)

```
mysql> USE DBMS;
Database changed
mysql> DELIMITER //
mysql>
mysql> CREATE FUNCTION build_string()
-> RETURNS VARCHAR(100)
-> DETERMINISTIC
-> BEGIN
->     DECLARE x INT DEFAULT 1;
->     DECLARE result VARCHAR(100) DEFAULT '';
->
->     WHILE x <= 5 DO
->         SET result = CONCAT(result, x);
->         SET x = x + 1;
->     END WHILE;
->
->     RETURN result;
-> END //
Query OK, 0 rows affected (1.438 sec)

mysql>
mysql> DELIMITER ;
mysql> SELECT build_string();
+-----+
| build_string() |
+-----+
| 12345          |
+-----+
1 row in set (0.135 sec)
```

2)

```

mysql> CREATE FUNCTION calculate_income()
-> RETURNS INT
-> DETERMINISTIC
-> BEGIN
->     DECLARE income INT DEFAULT 1000;
->
->     REPEAT
->         SET income = income + 1000;
->     UNTIL income >= 4000
->     END REPEAT;
->
->     RETURN income;
-> END //

```

Query OK, 0 rows affected (0.186 sec)

```

mysql>
mysql> DELIMITER ;
mysql> SELECT calculate_income();

```

```

+-----+
| calculate_income() |
+-----+
|                4000 |
+-----+
1 row in set (0.233 sec)

```

Ex:No: 13

Simple programming using CASE and LOOP

1)

```
MySQL 9.6 Command Line Cli  X  +  v
-> DETERMINISTIC
-> BEGIN
->     DECLARE income VARCHAR(20);
->
->     CASE
->         WHEN monthly_value <= 4000 THEN
->             SET income = 'Low Income';
->         WHEN monthly_value <= 5000 THEN
->             SET income = 'Avg Income';
->         ELSE
->             SET income = 'High Income';
->     END CASE;
->
->     RETURN income;
-> END //
```

ERROR 1304 (42000): FUNCTION income_level already exists

```
mysql>
mysql> DELIMITER ;
mysql>
mysql> SELECT income_level(3500) AS Income_Level;
+-----+
| Income_Level |
+-----+
| Low Income   |
+-----+
1 row in set (0.003 sec)
```

```
mysql> SELECT income_level(4500) AS Income_Level;
+-----+
| Income_Level |
+-----+
| Avg Income   |
+-----+
1 row in set (0.007 sec)
```

```
mysql> SELECT income_level(6000) AS Income_Level;
+-----+
| Income_Level |
+-----+
| High Income  |
+-----+
```

2)

```

mysql> DELIMITER //
mysql>
mysql> CREATE FUNCTION check_income()
-> RETURNS VARCHAR(30)
-> DETERMINISTIC
-> BEGIN
->     DECLARE income INT DEFAULT 1000;
->     DECLARE result VARCHAR(30) DEFAULT '';
->
->     myloop: LOOP
->         SET income = income + 1000;
->
->         IF income < 4000 THEN
->             ITERATE myloop;
->         END IF;
->
->         SET result = 'Income reached 4000';
->         LEAVE myloop;
->     END LOOP;
->
->     RETURN result;
-> END //

```

Query OK, 0 rows affected (0.109 sec)

```

mysql>
mysql> DELIMITER ;
mysql>
mysql> SELECT check_income() AS Result;

```

```

+-----+
| Result          |
+-----+
| Income reached 4000 |
+-----+

```

Ex:No: 14

TCL COMMANDS – COMMIT, SAVEPOINT, ROLLBACK

1)

```
mysql>
mysql> CREATE TABLE Class (
->     id INT PRIMARY KEY,
->     name VARCHAR(30)
-> );
Query OK, 0 rows affected (0.980 sec)

mysql>
mysql> INSERT INTO Class VALUES
-> (1, 'alpha'),
-> (2, 'beta'),
-> (3, 'gamma'),
-> (4, 'delta'),
-> (5, 'echo');
Query OK, 5 rows affected (0.161 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.005 sec)

mysql>
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.003 sec)

mysql>
mysql> UPDATE Class SET name='bravo' WHERE id=5;
Query OK, 1 row affected (0.061 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SAVEPOINT A;
Query OK, 0 rows affected (0.007 sec)

mysql>
mysql> INSERT INTO Class VALUES(6, 'uppal');
Query OK, 1 row affected (0.004 sec)

mysql> SAVEPOINT B;
Query OK, 0 rows affected (0.003 sec)
```

2)

```
mysql>
mysql> INSERT INTO Class VALUES(7,'balu');
Query OK, 1 row affected (0.004 sec)
```

```
mysql> SAVEPOINT C;
Query OK, 0 rows affected (0.003 sec)
```

```
mysql>
mysql> SELECT * FROM Class;
```

id	name
1	alpha
2	beta
3	gamma
4	delta
5	bravo
6	uppal
7	balu

```
7 rows in set (0.015 sec)
```

```
mysql>
mysql> ROLLBACK TO B;
Query OK, 0 rows affected (0.011 sec)
```

```
mysql>
mysql> SELECT * FROM Class;
```

id	name
1	alpha
2	beta
3	gamma
4	delta
5	bravo
6	uppal

```
6 rows in set (0.005 sec)
```


Ex:No: 15

DCL COMMANDS- GRANT, REVOKE

1)

```
mysql> CREATE TABLE Student (  
-> ID INT PRIMARY KEY,  
-> Name VARCHAR(30)  
-> );  
Query OK, 0 rows affected (0.841 sec)  
  
mysql>  
mysql> INSERT INTO Student VALUES (1, 'Sandra');  
Query OK, 1 row affected (0.120 sec)  
  
mysql>  
mysql> CREATE USER 'student_user'@'localhost'  
-> IDENTIFIED BY '1234';  
Query OK, 0 rows affected (0.569 sec)  
  
mysql> GRANT SELECT ON experiment15.Student  
-> TO 'student_user'@'localhost';  
Query OK, 0 rows affected (0.095 sec)  
  
mysql> SHOW GRANTS FOR 'student_user'@'localhost';  
+-----+  
| Grants for student_user@localhost |  
+-----+  
| GRANT USAGE ON *.* TO 'student_user'@'localhost' |  
| GRANT SELECT ON `experiment15`.`student` TO 'student_user'@'localhost' |  
+-----+  
2 rows in set (0.032 sec)
```

2)

```
mysql> GRANT INSERT ON experiment15.Student  
-> TO 'student_user'@'localhost';  
Query OK, 0 rows affected (0.053 sec)  
  
mysql> REVOKE INSERT ON experiment15.Student  
-> FROM 'student_user'@'localhost';  
Query OK, 0 rows affected (0.131 sec)  
  
mysql> SHOW GRANTS FOR 'student_user'@'localhost';  
+-----+  
| Grants for student_user@localhost |  
+-----+  
| GRANT USAGE ON *.* TO 'student_user'@'localhost' |  
| GRANT SELECT ON `experiment15`.`student` TO 'student_user'@'localhost' |  
+-----+  
2 rows in set (0.005 sec)
```